

Eye-Gaze Communication App for ALS Patients

User Manual

Assistive Technology for ALS Patients

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1. Overview

This app enables ALS (Amyotrophic Lateral Sclerosis) patients who have lost the ability to speak or move to communicate using only their eye movements. Using an Android phone (Galaxy phone) or tablet (Galaxy Tablet), the standard front-facing camera — no special hardware required — detects left and right gaze to build a binary codeword (e.g. 01). The codeword is matched to a word (e.g. “No”) or phrase, which is then spoken aloud through the device speaker via text-to-speech.

Unlike conventional gaze-tracking communication devices, which reconstruct words letter-by-letter from an on-screen virtual keyboard, this tool is designed around the small set of words a patient uses most. By selecting whole words with only a few gaze movements, the patient can communicate with far less fatigue.

1.1 Key Features

- Hands-free, hardware-free operation using the device's built-in camera
- Hierarchical word system covering 70+ words across 5 categories
- Automatic Zone 3 shortcut that adapts to the patient's most-used words
- Category pages for Pain, Action, Environ-State, and Feeling
- Text-to-speech output when a word is selected
- Eye-closure pause/resume for breaks
- Customizable vocabulary via the Setup screen
- Optimized layouts for Galaxy Phone and Galaxy Tab
- Full bilingual support: Korean / English — selectable on first launch

1.2 How It Works — Left/Right Gaze Tracking

When the app starts, the area of the patient’s face around the eyes appears in the front-camera preview, and the app begins detecting changes in the patient’s gaze. The patient communicates by turning the eyes left or right to build a binary sequence (see the table below). The movement is not a glance at the left or right side of the screen: the eyeballs must be turned as far as they will go, as if looking sideways at something past the edge of the device. A full turn to the left adds a 0; a full turn to the right adds a 1. When the same bit repeats — 00, for example — turn the eyes to that side, return briefly to the centre, then turn to the same side again. After completing the sequence, gazing at the center for 3 seconds (the default; adjustable from 2 to 5 seconds in Setup) confirms the selection.

Gaze direction	Bit added	Example
Look LEFT	0	Left = 0 = “Can't breathe”
Look RIGHT	1	Right = 1 = “Suction” (emergency)
Center gaze (3 s)	Confirm	Hold gaze at center for 3 seconds to confirm

NOTE: For best accuracy, remove glasses before use. If glasses must be worn, adjust the device or head angle slightly until lens reflections are no longer visible in the camera preview — reflections interfere with iris detection and reduce accuracy.

For the word “Uncomfortable,” the codeword is 011 (left–right–right), entered as LEFT → RIGHT → centre (briefly) → RIGHT. The short return to the centre separates the two consecutive RIGHT turns; when the direction changes, no pause is needed. The following images show a person generating this sequence (left/right are from the user's perspective). To complete the codeword, the patient gazes at the center for more than 3 seconds at the end.



CENTER (SHORT)

CENTER (3 SEC)

As shown in Figure 1, mount the phone or tablet roughly 30 cm from the patient's face — positioned so it does not block the patient's line of sight, while still capturing the patient's eye movements clearly. On a phone, use portrait: the camera sits at the top centre of the screen, so the left and right gaze directions are detected evenly. On a tablet, use landscape for the same reason. The whole face need not be in frame — the eye region is enough.

A woman is lying in bed, looking up at a tablet mounted on a stand. The tablet displays a Korean medical app interface. At the top, it says "환자정보 입력" (Patient Information Input) and "환자 정보 입력을 완료하십시오" (Please complete patient information input). Below this is a table with columns for "성명" (Name), "성별" (Gender), "나이" (Age), "진단" (Diagnosis), and "처방" (Prescription). The table has four rows of data. At the bottom, there are buttons for "시작" (Start), "환자정보" (Patient Information), and "진단" (Diagnosis).

성명	성별	나이	진단	처방
김민준	남	001	사상감	
이민준	남	010	심부전	
박민준	남	011	복합성	
정민준	남	012	복합성	

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2. Getting Started

2.1 Device Setup

- Supported devices: Samsung Galaxy Phone and Galaxy Tab (Android)
- Mount the device so the front camera is at eye level for the patient
- Ensure the room is well-lit — natural or overhead light works best
- Clean the camera lens before use
- Use without glasses, or adjust the device or head angle to eliminate lens glare/reflections

2.2 Device Type Selection

On the Instructions screen — the second screen — select the correct device type:

Device	Typical camera position
Galaxy Phone	Portrait: camera at top-center Landscape: camera at left-center
Galaxy Tab	Portrait: camera at right-center Landscape: camera at top-center

NOTE: *The patient must look toward the phone/tablet camera for gaze to be recognised reliably.*

On the Instructions screen, Setup opens the menu for editing the codeword phrases, and Start begins using the app.

2.3 First Launch — Language & Disclaimer / Instruction Page

On first launch a safety disclaimer is displayed (Figure 2A). At the top of the screen tap English or 한국어 (Korean) to choose the interface language — this is remembered across sessions. Read the disclaimer and tap Accept to continue. This app must always be used under caregiver supervision.

Button	Effect
English	Sets all UI text, word dictionary, and TTS to English
한국어(Korean)	Sets all UI text, word dictionary, and TTS to Korean

The Instructions screen then appears, as shown in Figure 2B. The Setup button opens the Setup screen (see Section 5, Setup Screen). The Practice button opens the practice screen, where the user practices building codewords with left and right gaze (see Section 2.7, Practice Mode). The Start button opens the main screen, where the user begins building the codeword for each phrase.

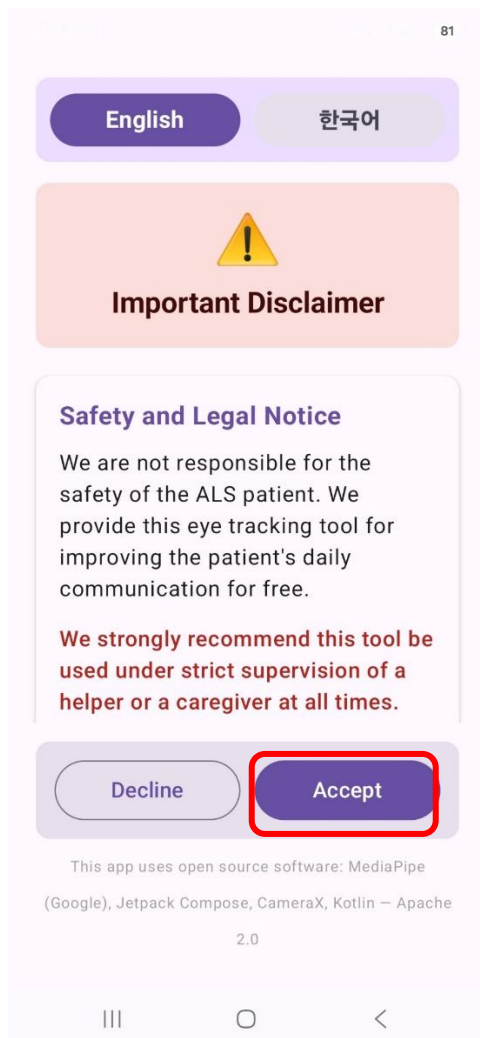


Figure 2A — Disclaimer page

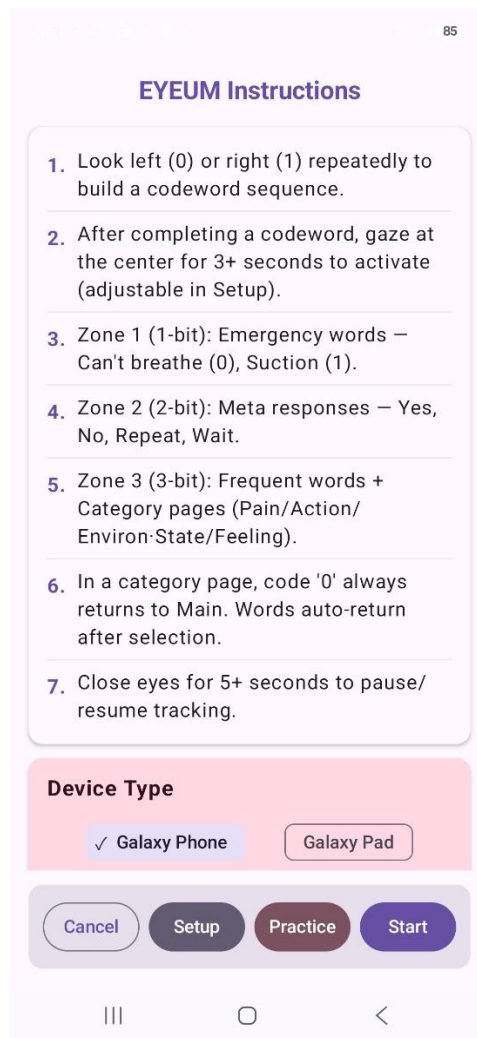


Figure 2B — Instruction page

2.4 Launching the Camera

On the main screen (Figures 3 and 4), tap Start. The camera activates, gaze tracking of both eyes begins, and the status indicator turns green (TRACKING). The blue dashed area shows the binary codeword as it is being built, and the red dashed area shows the words (and categories) with their corresponding codes.



Figure 3 — App main page (Landscape mode)



Figure 4 — App main page (Portrait mode)

Button	Function
Start	Activates camera and begins eye tracking
Pause/Resume	Temporarily suspends tracking (camera stays on). Press again to resume. The same pause/resume action can be triggered by closing the eyes for 5 seconds — so a patient who cannot press buttons can pause by closing their eyes for 5+ seconds during use, and resume by closing them again for 5+ seconds.
Stop	Deactivates camera and stops all tracking

2.5 Pausing with Eye Closure

Keeping the eyes closed for 5 or more seconds pauses or resumes tracking without caregiver assistance. The status indicator turns orange (PAUSED). Closing the eyes again for 5 seconds resumes tracking. Use this while resting or sleeping, or while listening to a visitor — any time the patient does not intend to enter a word. While paused, ordinary eye movements are not read as input.

2.6 Building Words with Left/Right Gaze (Important)

- The words an ALS patient can use are listed in the main window (shown by the red dashed outline in Figures 3 and 4). The patient identifies the available words and the codeword that corresponds to each one. For example, the codeword for “Can't breathe” is “0”.
- Each word is produced by building its codeword through left/right gaze. For example, gazing left then right once each produces the codeword “01”, which corresponds to the word “No”.
- When the same digit repeats in a codeword — 00 or 000, for example — turn the eyes to that side, return briefly to the centre, then turn to the same side again; otherwise the two gazes are read as one. No pause is needed when the direction changes.
- As the codeword is built, each gaze adds one digit shown in the blue dashed area of Figures 3/4, so the patient can watch the codeword being formed in real time.
- After the left/right gaze sequence, gazing at the center for 2 or more seconds completes the codeword. If the completed codeword exists in the word book, it is spoken aloud.
- If an incorrect codeword is formed, gazing at the center for 3+ seconds immediately will discard it automatically when the codeword does not exist in the word book; after 3 seconds the patient can build a new word with left/right gaze. If the mistakenly-formed codeword does happen to exist in the word book, it will be spoken — this is unavoidable.

2.7 Practice Mode

The first time the app is opened, a practice mode starts automatically in place of the phrase chart. Press Start and follow the voice guidance. The practice has three steps, and each step is passed after two correct entries in a row for every target.

Step 1 — Turn the eyes fully to one side (0 or 1). The direction is judged as soon as it is detected; no centre hold is needed.

Step 2 — Enter the same side twice (00 or 11), passing the centre briefly in between, then look at the centre to confirm.

Step 3 — Enter three moves that change direction (010 or 101), then look at the centre to confirm.

After each attempt, the entered code is shown in green with ✓ when correct, or in red with X together with a spoken hint — for example, that the eyes should be turned further, or that the centre was held too long between repeated digits. The next target appears after the eyes have rested at the centre for about one second.

Practice can be repeated at any time with the Practice button on the instructions screen, and a caregiver can tap Skip practice to start using the app immediately. When practice is complete, the screen shows the average response time and the number of errors for each direction. Please include these numbers when reporting problems with left/right detection.

3. The Codeword System

Words are organized into four zones based on code length (the differently-colored regions [red, green, yellow, blue] in Figure 5) plus four category sub-pages. By design, a single screen can show only 12 words (as in Figure 3). To fit more words into the word book, Zone 3B (blue) is used to select sub-categories: when a Zone 3B codeword is chosen, the app opens the corresponding category sub-page.



Figure 5 — App main page (zone colour map)

3.1 Zone Overview

Zone	Code length	Gaze actions	Purpose
Zone 1	1 bit	1 + center	Emergency words — fastest access with a single left/right gaze
Zone 2	2 bits	2 + center	Meta responses (Yes / No / Repeat / Wait) — selected with two gazes
Zone 3A	3 bits (000–011)	3 + center	4 most-used words — auto-updated by frequency, selected with three gazes
Zone 3B	3 bits (100–111)	3 + center	Category triggers (Pain / Action / Environ·State / Feeling), selected with three gazes
Sub-pages	1–3 bits	3+center, then 1–3+center	Up to 13 words per category (code 0 = back to main)

3.2 Main Page — Full Word List

Code	Zone	Gaze steps	Phrase
0	Zone 1 ●	1 + center	Can't breathe
1	Zone 1 ●	1 + center	Suction
00	Zone 2 ☑	2 + center	Yes
01	Zone 2 ☑	2 + center	No
10	Zone 2 ☑	2 + center	Repeat
11	Zone 2 ☑	2 + center	Wait
000	Zone 3 ●	3 + center	Thirsty ★
001	Zone 3 ●	3 + center	Urinate ★
010	Zone 3 ●	3 + center	Defecate ★
011	Zone 3 ●	3 + center	Uncomfortable ★
100	Zone 3 ●	3 + center	→ PAIN page
101	Zone 3 ●	3 + center	→ ACTION page
110	Zone 3 ●	3 + center	→ ENVIRON·STATE page
111	Zone 3 ●	3 + center	→ FEELING page

NOTE: ★ Zone 3 shortcuts (000–011) automatically update to show the 4 most-frequently used category words. See Section 4.

NOTE: For codes with repeated digits (00, 000), the eyes must return briefly to the centre between the repeats, so the actual number of gaze movements is higher than the Gaze steps column shows.

3.3 Category Pages

Entering a category page resets the code to – (empty). Code 0 (look left + confirm) always returns immediately to main. After any word is selected the app automatically returns to main.

PAIN page (enter with code 100)

Code	Phrase
0	← Back to Main
1	Stomachache
00	Headache
01	Back pain
10	Neck pain
11	Chest pain
000	Nausea
001	Mouth dry
010	Eyes dry
011	Leg pain
100	Shoulder pain
101	Cramp
110	Numb
111	Getting worse

ACTION page (enter with code 101)

Code	Phrase
0	← Back to Main
1	Medicine
00	Change posture
01	Turn me
10	Lift head
11	Lower head
000	Wipe
001	Cover me
010	Call nurse
011	Come here
100	Sit me down
101	Hold me still
110	Urinate
111	Defecate

ENVIRON-STATE page (enter with code 110)

Code	Phrase
0	← Back to Main
1	Cold
00	Hot
01	Too bright
10	Too dark
11	Too loud
000	Sleepy
001	Itchy
010	Thirsty
011	Uncomfortable
100	Open window
101	Close window
110	Turn on TV
111	Turn off TV

FEELING page (enter with code 111)

Code	Phrase
0	← Back to Main
1	Thank you
00	Sorry
01	Anxious
10	Scared
11	Sad
000	Lonely
001	Happy
010	Exhausted
011	Angry
100	Miss you
101	I love you
110	Proud of you
111	Don't worry

4. Zone 3A Adaptive Shortcut

The four Zone 3A frequency slots (codes 000, 001, 010, 011) automatically update to show the patient's most-used words from any category page.

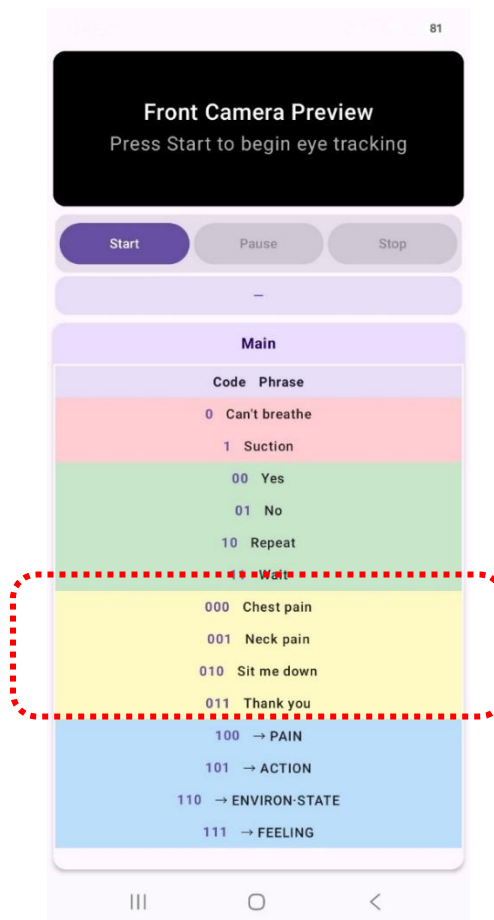


Figure 6 — Zone 3A shortcut display

4.1 How Frequency Tracking Works

- Each time a word is selected from a category page, its usage count increases by 1
- Counts are saved to the device and persist between sessions
- The Zone 3A slots are recalculated only when Stop is pressed or the app is restarted, never during a session, so a slot cannot change meaning between reading the board and finishing a code
- The four highest-count words appear in Zone 3A in descending order
- If fewer than 4 words have been used, defaults fill the remaining slots: Thirsty, Urinate, Defecate, Uncomfortable. These four remain reachable from the ACTION and ENVIRON-STATE pages even when other words occupy Zone 3A

4.2 Viewing and Resetting Counts

Open Setup and scroll to the yellow Zone 3A — Usage Frequency panel. Tap Show to expand: this displays the current Zone 3A shortcuts with their usage counts and the overall top 10 words. To reset all counts, tap Reset all usage counts.

5. Setup Screen

The Setup screen allows caregivers to customize the vocabulary — words can be added to or removed from the word book. To add or remove words, tap Setup on the Instructions screen (Figures 7 and 8).

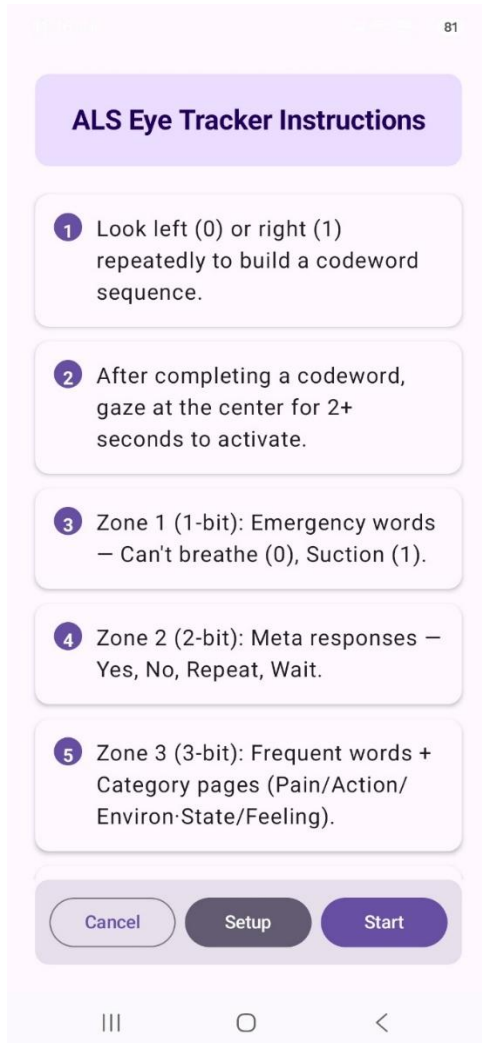


Figure 7 — Instruction screen — before entering Setup

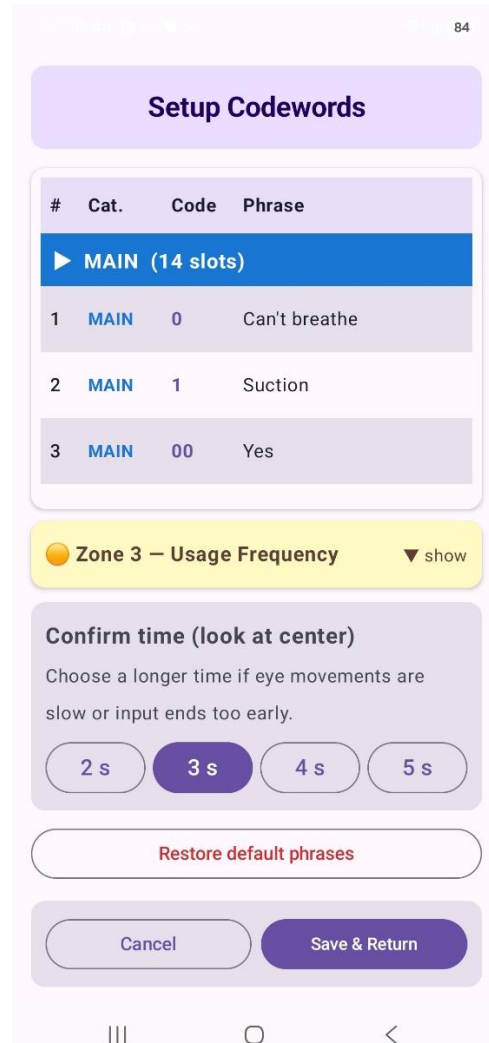


Figure 8 — Setup page — after entering Setup

5.1 Navigating the Word List

Words are grouped by category with color-coded headers:

Color	Category	Contains
Blue	MAIN	Emergency, meta, Zone 3 shortcuts, category triggers
Red	PAIN	14 pain / symptom slots
Dark blue	ACTION	14 caregiver action slots
Green	ENVIRON·STATE	14 environment / condition slots
Orange	FEELING	14 emotional expression slots

Each category has exactly 14 slots for codes 0, 1, 00, 01, 10, 11, 000, 001, 010, 011, 100, 101, 110, 111. Empty slots show *** in grey.

5.2 Editing a Word

- Tap any row to open the Edit dialog
- Type the new phrase and tap Save
- To clear a word back to ***, tap Clear (set to ***)
- Tapping a slot shows the edit dialog — type the phrase and tap Save. To undo every edit at once, tap Restore default phrases at the bottom of the Setup screen
- Tap Save & Return to write all changes to storage

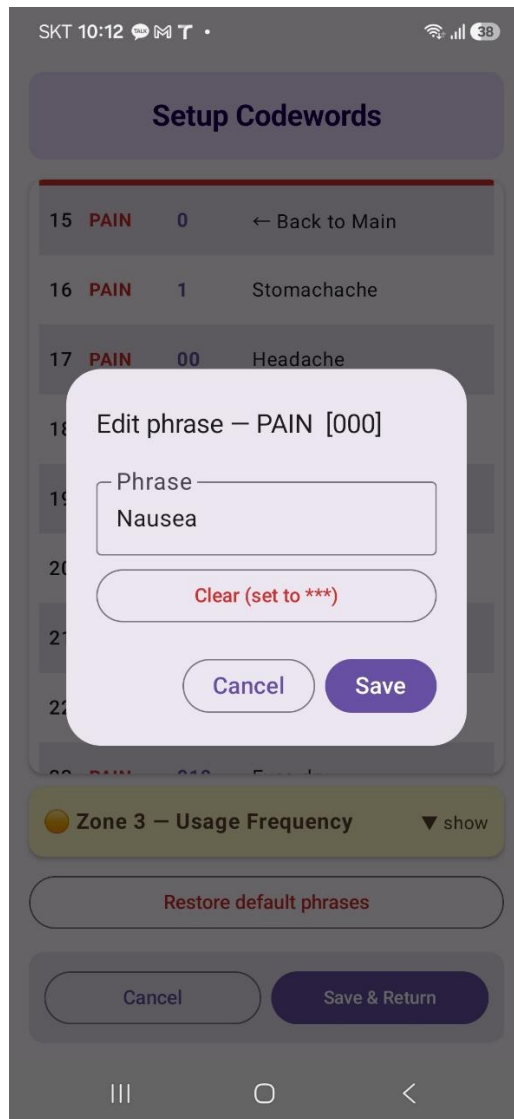


Figure 9 — Add-word dialog popup after tapping a row

NOTE: *Changes take effect on the main screen immediately. Closing with Cancel discards all changes.*

5.3 Words That Cannot Be Edited

- Code 0 on main page: Can't breathe (emergency — always fastest access)
- Code 0 on any category page: Back to Main (navigation — essential)
- Codes 100–111 on main page: category page triggers

5.4 Confirm Time

The confirm time is how long the patient must look at the centre to complete a codeword. It can be set to 2, 3, 4, or 5 seconds in the Confirm time section of the Setup screen; the default is 3 seconds. The change takes effect immediately and is kept after the app is closed.

Choose a longer time if the patient's eye movements are slow, or if an entry is completed too early when the same digit is repeated (for example, 00), because the brief return to the centre between the two digits must be shorter than the confirm time. A shorter time can be chosen once the patient is experienced. The instructions screen and the practice mode always show the time currently selected.

6. Screen Layout

6.1 Galaxy Phone — Landscape

Left region	Right region
Camera preview (left 1/3), Start / Pause / Stop buttons	Status bar, Code table (right 2/3, two-column)

6.2 Galaxy Phone — Portrait

Top to bottom:

- Camera preview (top)
- Start / Pause / Stop buttons
- Status bar
- Code table (portrait, most of screen)

6.3 Galaxy Tab — Landscape

- Camera preview (top)
- Status bar + Code table (bottom, large text, two-column centered)
- Start / Pause / Stop buttons (bottom)

6.4 Galaxy Tab — Portrait

- Camera preview (top)
- Start / Pause / Stop buttons
- Status bar
- Code table (large text, centered, most of screen)

NOTE: In Galaxy Tab portrait mode, the front camera can be far from the camera view, which may reduce recognition accuracy. Landscape mode is recommended for the Galaxy Tab.

On Galaxy Tab, text is enlarged approximately 1.7× and aligned toward the screen center. In landscape mode the left column is right-justified and the right column is left-justified so content converges toward the center, minimizing unnecessary left-right eye movement.

7. Step-by-Step Examples

7.1 Saying “Can't breathe” (emergency)

Step	Action
1	Look LEFT (adds bit 0)
2	Gaze at center 3 s → “Can't breathe” is spoken

7.2 Saying “Headache” via PAIN page

Step	Action
1	Look RIGHT → LEFT → centre (briefly) → LEFT (builds code 100 — PAIN trigger)
2	Gaze center 3 s → PAIN page opens
3	Look LEFT → centre (briefly) → LEFT (builds sub-code 00)
4	Gaze center 3 s → “Headache” spoken → app returns to Main

7.3 Returning to Main page from a category

Step	Action
1	Look LEFT (builds sub-code 0)
2	Gaze center 3 s → returns to Main (no speech output)

7.4 Saying “Thank you”

Step	Action
1	Look RIGHT → centre (briefly) → RIGHT → centre (briefly) → RIGHT (builds code 111 — FEELING trigger)
2	Gaze center 3 s → FEELING page opens
3	Look RIGHT (builds sub-code 1)
4	Gaze center 3 s → “Thank you” spoken → app returns to Main

8. Tips and Best Practices

8.1 Camera Positioning

- Camera should be at the same height as the patient's eyes
- Keep the patient's eye region within the center of the camera frame
- Avoid back-lighting (e.g. a window behind the patient) — position light sources in front of the patient
- Distance: approximately 30 cm
- To avoid blocking the patient's line of sight, mount the phone or tablet slightly above eye/face level

Eyeglasses: For best tracking accuracy, remove glasses before use if possible. If glasses must be worn, adjust the device or head angle slightly until reflections (glare) from the lenses are no longer visible in the camera preview. Reflections interfere with iris detection and reduce accuracy.

8.2 Training New Users

- Begin with the built-in practice mode (Practice button on the instructions screen) before using the phrase chart
- Start with Zone 1 and Zone 2 (1–2 bit codes) and practice until consistent
- Introduce Zone 3A shortcuts once Zones 1 and 2 are reliable
- Introduce category pages one at a time
- Short sessions (10–15 minutes) are better than long ones when starting out

8.3 Optimising Zone 3A

- Use category pages regularly — Zone 3A adapts automatically after real use
- Review the Usage Frequency panel weekly via Setup → Zone 3A — Usage Frequency

8.4 Adding Custom Words

- Open Setup and tap any *** slot to add a custom phrase
- Keep phrases short (1–3 words) for clear TTS output

8.5 Gaze Sensitivity

If the system is not registering gaze correctly:

- Increase room lighting
- Ensure the patient's eyes are fully visible (no drooping eyelids)
- Wipe the camera lens
- Check that the device has not been moved from its original position

9. Troubleshooting

Problem	Solution
Camera does not start	Grant camera permission: Android Settings → Apps → EYEUM → Permissions
No speech output	Check device volume (physical volume buttons). Verify the TTS engine: Android Settings → General Management → Language & Input → Text-to-Speech
Gaze not detected	Improve lighting. Clean lens. Move patient closer. Remove glasses if possible; if not, adjust device angle to eliminate lens reflections
Tracking too sensitive	Make sure the eyelids are not half-closed. Ordinary small eye movements should not enter a code
Zone 3 shows unexpected words	Setup → Zone 3 — Usage Frequency → Reset all usage counts to restore defaults
Words shown in wrong language	Return to the Disclaimer screen and re-select the language to reload the correct default dictionary

10. Safety Notice

This application is an assistive communication aid only. It is not a medical device and must not be used as the sole means of alerting caregivers to emergencies. Always ensure a caregiver is present and monitoring the patient during use.

- The app must be used under caregiver supervision at all times
- Do not rely solely on this app for emergency communication — always have a backup alert method
- The developers are not liable for any consequences arising from use of this application
- Obtain a speech-language pathology evaluation before introducing this system for extended use

By using this application, you acknowledge and accept all risks associated with its use.

Questions, problems, or feedback: contact.eyyum@gmail.com

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